

Question	Answer	Marks
1(a)(i)	Loss in gravitational potential energy $\Delta GPE = mg\Delta h = (0.400) (9.81) (0.200) = \mathbf{0.785 \text{ J}}$	1
(a)(ii)	<u>Method 1:</u> By Newton's second law, taking upwards as positive, $F_{\text{net}} = m a$ $F_{\text{spring}} - W = 0$ $k x = m g$ $k = m g = (0.400 \times 9.81) / (0.200) = \mathbf{19.62 \text{ N m}^{-1}}$ The elastic potential energy stored $\Delta EPE = \frac{1}{2} k x^2 = \frac{1}{2} (19.62) (0.200)^2 = \mathbf{0.392 \text{ J}}$ <u>Method 2:</u> In equilibrium, the spring force is equal to the weight of the object, $F_{\text{spring}} = W$ $k x = m g$ $k = m g = (0.400 \times 9.81) / (0.200) = \mathbf{19.62 \text{ N m}^{-1}}$ The elastic potential energy stored $\Delta EPE = \frac{1}{2} k x^2 = \frac{1}{2} (19.62) (0.200)^2 = \mathbf{0.392 \text{ J}}$	1 1 1 1
(b)	The difference is work done against the external force needed to support the mass while lowering it gently. This force is the difference between the mass's weight and the tension in the spring. OR A variable external upward force is required when the spring mass is stretched gently downwards towards its equilibrium point. Thus some gravitational potential energy is lost due to the negative work done by the external force, while the remainder is converted to elastic potential energy.	1 1
(c)(i)	Angular frequency of mass-spring system $\omega = \sqrt{k/m} = \sqrt{19.62/0.400} = \mathbf{7.00 \text{ rad s}^{-1}}$ Maximum speed $v_0 = \omega x_0 = 7.00 \times 0.100 = \mathbf{0.700 \text{ m s}^{-1}}$	1 1
(c)(ii) & (iii)	Both P and Q are labelled correctly (indicated with a dot or a cross) Correct axes labels of v_{max} and amplitude. A curve starting at a displacement of -0.100 m and spiralling with smaller v and x in clockwise direction.	1 1 1

	Max Marks	10

Question	Answer	Marks
2(c)	The total linear momentum of a system is CONSTANT if no net external force acts	1